

SESSION 5

RAD-423 Imaging Informatics

Health Data Analytics, Interoperability & Applied Practice

Using imaging data, interoperability and AI to improve clinical and operational workflow

Course	RAD-423 Imaging Informatics
Date	Sept, 2026
Lecturer	A. A. Zubair
Suggested study time	60–90 minutes

Learning Objectives

By the end of this session, the student should be able to:

1. define health data analytics and identify common imaging data sources used for analysis;
2. distinguish descriptive, diagnostic, predictive and prescriptive analytics;
3. explain common radiology workflow indicators such as turnaround time, waiting time, scanner utilization and no-show rate;
4. explain why data quality and governance are essential for reliable analytics;
5. define interoperability and describe the broad roles of DICOM, HL7 and FHIR;
6. explain how artificial intelligence can support imaging analysis and workflow; and
7. apply imaging-informatics concepts to an acute-stroke workflow and the radiographer's role.

Key Concept

Imaging data become useful only when they are accurate, interpretable and connected to the clinical workflow. Analytics helps convert data into information; interoperability allows systems to exchange and use that information; people then apply it to clinical and operational decisions.

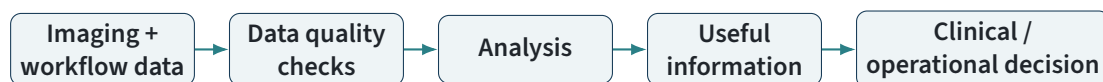
1 Health Data Analytics in Medical Imaging

health data analytics is the systematic analysis of healthcare data to generate useful information for clinical, operational, research and administrative decision-making.

In imaging departments, data are produced at nearly every stage of the patient journey. The final diagnostic image is only one component. Other useful data may include:

- patient and demographic information;
- examination type and protocol;
- imaging and derived measurements;
- radiology reports;
- acquisition and radiation-dose information;
- scanner and equipment information;
- scheduling and appointment data;
- examination duration and waiting time;
- reporting turnaround time; and
- operational events such as scanner downtime or missed appointments.

The purpose of analytics is not merely to create graphs or dashboards. A useful analysis begins with a defined question, uses appropriate data, and supports a clinical or operational decision.



Common Error

A large dataset does not automatically produce reliable knowledge. If patient identifiers, examination codes, dates or workflow timestamps are incomplete or incorrect, sophisticated analysis can still produce misleading results.

2 Four Levels of Analytics

Analytics are grouped into four levels. These are best understood by the **question** each level asks.

2.1 1. Descriptive analytics: What happened?

Descriptive analytics summarizes events that have already occurred. It is the starting point for most departmental reporting.

Examples include:

- How many MRI examinations were performed last month?
- What proportion of CT examinations used contrast?
- What was the median report turnaround time last week?
- Which examination type had the greatest volume?

The answer describes the current or historical situation but does not, by itself, explain the cause.

2.2 2. Diagnostic analytics: Why did it happen?

Diagnostic analytics investigates factors associated with an observed result. For example, if MRI waiting time increases, possible contributors might include:

- increased patient volume;
- staff shortage;
- scanner downtime; or
- longer examination protocols.

This stage often requires comparing groups, time periods or workflow variables rather than simply counting events.

2.3 3. Predictive analytics: What is likely to happen?

Predictive analytics uses existing data patterns to estimate a future event or outcome. For example, a predictive model might estimate which patients are at increased risk of missing an MRI appointment.

The output is not certainty. A prediction is an estimate based on the available data, the quality of those data and the method used.

2.4 4. Prescriptive analytics: What should we do?

Prescriptive analytics goes beyond prediction and asks which action may help achieve a desired objective. For example, appointment scheduling might be redesigned to reduce waiting time or improve scanner use.

Analytics type	Core question	Imaging example
Descriptive	What happened?	Count MRI examinations performed during the month
Diagnostic	Why did it happen?	Investigate why waiting time increased
Predictive	What is likely to happen?	Estimate the risk of appointment non-attendance
Prescriptive	What should we do?	Recommend scheduling changes to reduce waiting time

Focus

Do not define the four analytics types only by listing their names. State the **question each type answers** and give an imaging-related example.

3 Worked Imaging-Analytics Example

Suppose a one-year MRI dataset containing variables such as patient age, sex, examination type, date, scan duration, protocol, contrast use, radiologist, reporting time, scanner and diagnosis.

From the same dataset, different questions can be asked:

Question	Likely analytics level
Which MRI examination is most frequently performed?	Descriptive
Which protocol has the longest average scan time?	Descriptive
Why did average scan time increase after a protocol change?	Diagnostic
Which patients are most likely to miss their appointments?	Predictive
How should appointment slots be allocated to reduce delays?	Prescriptive

Example: investigating long MRI waiting times

Suppose the average waiting time has increased. A descriptive analysis confirms that the delay is real. Diagnostic analysis then compares patient volume, staffing, scanner downtime and protocol duration. If long protocols and repeated downtime are major contributors, management decisions should address those causes rather than simply asking staff to “work faster.”

4 Radiology Workflow Analytics

A simplified radiology workflow is:



Each transition can generate measurable timestamps or events. These allow a department to quantify performance rather than relying only on impressions.

4.1 Turnaround time

Turnaround time (TAT) is defined as the time from examination completion to the finalized report. A long TAT may indicate a reporting bottleneck, but the value should always be interpreted in context because different examination types and urgency categories may require different expectations.

4.2 Waiting time

Waiting time is the interval between appointment/request and examination. A waiting-time analysis can help reveal scheduling pressure, limited capacity or process delays.

4.3 Scanner utilization

Scanner utilization represents the proportion of available scanner time being used. Extremely low utilization may indicate unused capacity, while very high utilization can leave little flexibility for emergencies, maintenance or delays.

4.4 No-show rate

The **no-show rate** is the proportion of patients who do not attend scheduled appointments. Non-attendance wastes reserved capacity and may delay care for both the absent patient and others waiting for appointments.

Clinical / Practical Link

A department can have excellent image quality but poor service performance. Imaging informatics allows clinical quality and operational quality to be examined together rather than treating the scanner as an isolated device.

5 Data Quality and Governance

A central principle of analytics is summarized as:

Garbage in → garbage out

Poor-quality source data can undermine every later step. Examples include:

- incorrect patient demographics;
- duplicate patient records;
- missing information;
- incorrect examination codes;
- inconsistent terminology; and
- poor documentation.

These are not merely administrative problems. They can distort workload statistics, misclassify examinations, separate relevant clinical information from the correct patient and compromise the validity of research or quality-improvement projects.

Data quality dimensions

A practical data-quality check asks whether the data are **accurate, complete, consistent, timely and correctly linked to the intended patient or examination**.

Common Error

Do not “clean” a dataset by silently guessing missing values or changing unfamiliar codes. Data-quality correction should preserve traceability and follow an agreed governance process.

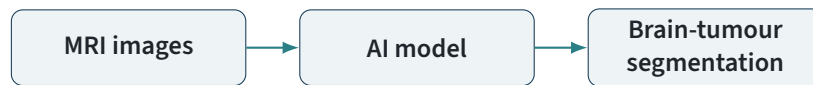
6 Artificial Intelligence and Imaging Analytics

Artificial intelligence can support:

- image classification;
- segmentation;
- detection;
- quantification;
- prognosis;
- workflow prioritization; and

- quality control.

A simple imaging example is:



The output may identify regions such as tumour core, enhancing tumour or peritumoral oedema. The important informatics point is that AI depends on **input data, software integration, output communication and human interpretation**. An algorithm that cannot receive the correct study or return its result to the appropriate workflow has limited clinical value.

High-Yield

AI is part of the imaging-informatics ecosystem, not a replacement for it. Reliable patient identity, image transfer, data quality, system integration and appropriate clinical review remain necessary.

7 Interoperability

Interoperability is defined as the ability of different healthcare systems to exchange and use information effectively.

This definition contains two parts:

1. systems must be able to **exchange** information; and
2. the receiving system must be able to **use** the information meaningfully.

Merely moving a file from one computer to another is therefore not enough if the receiving system cannot understand what the data represent.

7.1 DICOM

In this lecture series, DICOM has been introduced as the major standard for medical-imaging information and communication. It supports the structured handling of images and associated imaging information.

7.2 HL7

HL7 is widely used for exchanging healthcare information. In a radiology environment, this may include patient, order, result or workflow information rather than image pixels themselves. In a radiology environment, healthcare-information exchange may involve patient, order, result or workflow information rather than image pixels themselves.

7.3 FHIR

FHIR (Fast Healthcare Interoperability Resources) is a modern standard for exchanging healthcare information using web-oriented resources and APIs.

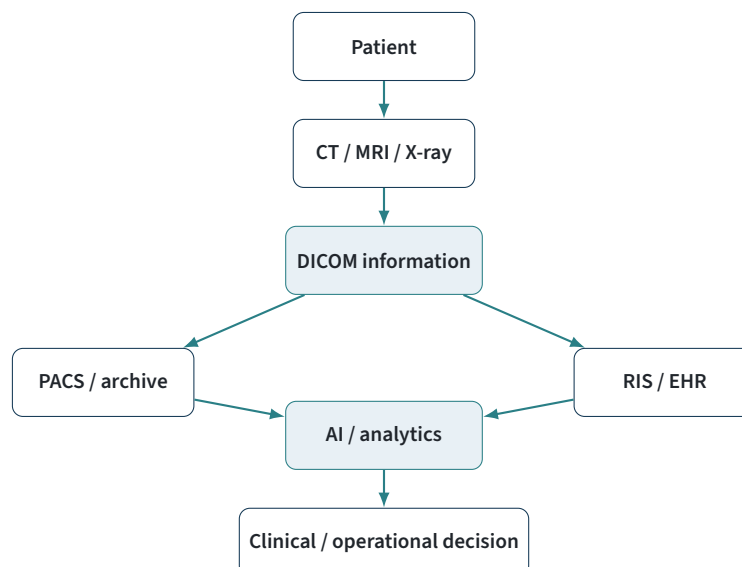
Standard	Primary emphasis in this lecture	Simple imaging-context view
DICOM	Medical imaging information	Images, imaging metadata and imaging communication
HL7	Healthcare information exchange	Patient, order, result and workflow information
FHIR	Modern healthcare information exchange via web-oriented resources/APIs	Structured clinical information that can be exchanged between applications

Common Error

Do not describe DICOM, HL7 and FHIR as competing versions of the same file format. They address overlapping but different parts of healthcare information exchange and may be used together in one clinical workflow.

8 The Integrated Imaging-Informatics Ecosystem

The simplified version below shows the relationship among acquisition, image management, workflow information, AI/analytics and clinical decision-making.

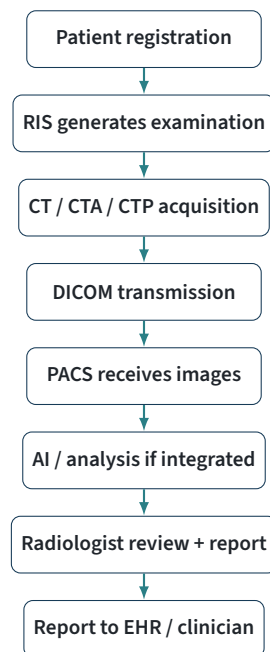


The value of the ecosystem lies in the connections. Images, workflow information and analytic outputs must reach the correct user, in the correct context, at the correct point in the clinical process.

9 Applied Case Study: Acute Stroke

A patient was brought to the imaging department with sudden neurological symptoms and suspected acute stroke. The patient undergoes CT, CT angiography and CT perfusion.

The informatics workflow is:



This case demonstrates why imaging informatics is clinically important. A delay or error at any stage can affect the time at which information reaches the treating team.

Where can the workflow fail?

Consider four possible failures: the wrong patient is selected at registration; the images fail to transmit to PACS; the AI service receives an incomplete study; or the report does not reach the clinician. In each case, the CT scanner may have produced technically adequate images, yet the overall imaging service has failed to deliver information correctly. Informatics therefore contributes directly to patient safety and timeliness.

10 Role of the Radiographer

The radiographer is not merely an image producer. Modern radiographers interact with PACS, RIS, EHR, DICOM, AI, quality-assurance systems, analytics and cybersecurity processes.

In practical terms, the radiographer contributes to:

Data accuracy

Correct patient identity, examination selection and documentation.

Image quality

Acquisition of diagnostically appropriate images and complete studies.

Data transfer

Verification that studies reach the correct PACS or downstream system.

Workflow quality

Correct completion and status of examinations within the departmental system.

Data security

Appropriate access and handling of patient information.

Analytics quality

Reliable timestamps, codes and workflow documentation that make later analysis meaningful.

Clinical / Practical Link

When a radiographer enters the wrong examination code, leaves a study incomplete in the RIS, or fails to notice a transmission error, the problem can propagate into reporting, billing, analytics, patient care, treatment planning and potentially patient harm. Good informatics practice begins at the point of data creation.

11 A Practical Analytics Workflow

A useful practical sequence helps connect imaging data with quality-improvement decisions. A basic workflow is:

1. **Define the question** — e.g., Why has MRI waiting time increased?
2. **Identify relevant data** — appointment dates, scan durations, staffing, downtime and protocol type.
3. **Check data quality** — identify missing, duplicate or inconsistent records.
4. **Analyze the data** — summarize, compare or model according to the question.
5. **Interpret the result** — distinguish what the data actually show from assumptions.
6. **Act and evaluate** — implement a justified change and measure whether the outcome improves.

Analytics should close the loop

Analytics is most useful when it leads to an action that is later re-measured. A dashboard that is never interpreted or acted upon is information display, not effective quality improvement.

Key Takeaways

- Health data analytics converts healthcare data into information for clinical, operational, research and administrative decisions.
- Descriptive, diagnostic, predictive and prescriptive analytics answer different questions and should not be used as interchangeable labels.
- Radiology workflow can be measured using indicators such as turnaround time, waiting time, scanner utilization and no-show rate.
- Poor source data can invalidate later analysis; accurate patient, examination and workflow information is therefore essential.
- AI can support image and workflow tasks, but it depends on reliable data and integration with the wider imaging-informatics ecosystem.
- Interoperability means systems can exchange and use information effectively.
- DICOM, HL7 and FHIR address different but complementary parts of healthcare information exchange.
- The acute-stroke example demonstrates how registration, RIS, acquisition, DICOM, PACS, AI, reporting and EHR communication form one time-sensitive clinical workflow.
- Radiographers influence data quality, transfer, workflow integrity and therefore the reliability of later analytics.

Key Terms

Health data analytics

Systematic analysis of healthcare data to support decision-making.

Descriptive analytics

Analysis focused on what happened.

Diagnostic analytics

Analysis focused on why an observed event or pattern occurred.

Predictive analytics

Analysis used to estimate what is likely to happen.

Prescriptive analytics

Analysis used to support decisions about what action should be taken.

Turnaround time

In this lecture, time from examination completion to finalized report.

Interoperability

Ability of different systems to exchange and use information effectively.

FHIR

Fast Healthcare Interoperability Resources; a modern standard for healthcare information exchange using web-oriented resources/APIs.

Self-Assessment Questions

1. Define health data analytics and list four types of imaging-related data that can be analyzed.
2. Distinguish descriptive, diagnostic, predictive and prescriptive analytics, giving one radiology example of each.
3. Define turnaround time, waiting time, scanner utilization and no-show rate.
4. Explain the statement “garbage in, garbage out” using an imaging-department example.
5. Define interoperability and explain the broad roles of DICOM, HL7 and FHIR.
6. Give four applications of AI identified in the lecture and explain why AI still depends on imaging informatics.
7. Using the acute-stroke case, trace the information pathway from patient registration to the clinician’s treatment decision.
8. A department discovers that MRI waiting time has increased. Outline a logical analytics workflow for investigating and responding to the problem.

